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Assessment of the stock status of small-scale and multi-gear fisheries resources in the tropical Eastern Pacific region

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ABSTRACT

Small-scale multi-gear fisheries contribute half of global fisheries landings but are generally data-poor, hindering their assessment and management. Aiming to overcome various existing challenges, we used two complementary length-based approaches to assess the status of three main target species in the small-scale fisheries of Eastern Pacific countries: Spotted rose snapper *Lutjanus guttatus*, Pacific sierra *Scomberomorus sierra*, and Pacific bearded *Brotula clarkae*, using length-frequency catch data (LFCD) from the Colombian Pacific coast. Two data sources – official governmental data and community-based monitoring from a non-government organization – were used to estimate two sets of stock indicators: one based on the derivation of growth and mortality parameters from modal progression, catch curve analysis and a yield-per-recruit model using TropFishR; and the second based on the relative contribution of fish sizes with regard to proposed reference values for healthy stocks. Growth estimates differed between data sources and exhibited large confidence intervals, indicating an overall high uncertainty underlying the LFCD revealed through a novel bootstrapped approach. Estimated values of stock indicators, exploitation rate, fishing mortality and size-proportions converged in suggesting a state of heavy to over-exploitation for the three assessed species, although differences were observed among data sources that we attribute mainly to fisheries selectivity and sampling design. In order to improve future assessments of stocks in multi-gear and data-poor contexts, estimations of fleet-specific selectivity should be used to reconstruct LFCD prior to analyses. Additionally, sampling design should be based on fishing effort distribution among gears and areas and, when feasible, fishery-independent data on stock conditions should be included.

1. Introduction

Small-scale fisheries are recognized as an "economic and social engine, providing food and nutrition security, employment and other multiplier effects" to local communities worldwide and contribute to nearly half of reported global fish catches (FAO, 2015). Moreover, when direct human consumption of fisheries products is included, the contribution of small-scale fisheries to global catch increases to two-thirds (FAO, 2015). This is particularly true for developing countries, where conditions of social inequity and poverty in rural communities increase the degree of dependence and the synergetic benefits derived from small-scale fisheries by coastal households (Béné et al., 2010).

Considering the socio-economic importance of small-scale fisheries, knowledge about the stock condition and future potential for exploitation of target species is essential for both users and managers of fishing resources. However, multiple sources of uncertainty impinge on traditional stock assessment methods (Hilborn and Walters, 1992; Scott et al., 2016). This uncertainty is amplified in data-poor contexts where unreported catches can lead to underestimation of the fishing impact on natural populations and to bias in temporal trends of total annual landings (Jacquet et al., 2010; Pauly and Zeller, 2016; Pitcher et al., 2002). Limited human and financial resources for fisheries management in developing countries often result in lack of continuity of fisheries statistics, lack of standardized data collection methods and under-representation of landing sites (Saavedra-Diaz, 2012; Salas et al., 2007; Zeller et al., 2006). In such contexts, incorporating data from participatory fisheries monitoring has the potential of improving the assessment of stock condition (Ramírez et al., 2017), by including less accessible

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landing sites and increasing the frequency of sampling and the overall sample sizes.

Small-scale fisheries of Colombia share many of the characteristics identified for this type of fisheries in Latin America: multi-gear and multi-species, low capital investment, labor intensive, remote and diverse landing sites and low management capacities from government authorities (Castellanos-Galindo and Zapata, 2018; Salas et al., 2007). In the Colombian Pacific coast, small-scale fisheries provide food, income and employment to at least 11,000 households of Afro-Colombian communities (Rueda et al., 2011). According to latest government fisheries statistics, they contribute 15%–40% of total landed catch in this coastal region, and this contribution increases to 68%–83% when landings of tuna species from the industrial fishing fleet are excluded (De la Hoz and Manjarrés-Martínez, 2016). Preliminary stock assessments, carried out as part of the national fisheries authority's duties to assess fisheries resources and establish annual catch quotas, indicate a status of over-exploitation of 50%–67% of main target species in the Colombian Pacific (Barreto and Borda, 2008; Barreto et al., 2010; Puentes et al., 2014). Additionally, a high proportion (>50%) of juvenile fish – below the length at maturity – are reported in the catch of most of those species (AUNAP and UNIMAGDALENA, 2013b), which could be a sign of growth and recruitment overfishing (Froese et al., 2008). However, these government reports acknowledge limitations in the data they had available for analysis, linked to the lack of funding to carry out continuous and systematic fisheries monitoring and to adequately quantify fishing effort. Besides the recurrent funding constraints, fisheries statistics in Colombia have also been affected by institutional changes that occurred between 2002 and 2011, in which fisheries management and enforcement responsibilities were transferred three times from one government authority to another (Saavedra-Diaz, 2012; Wielgus et al., 2010).

In this study, we carry out an assessment of the stock condition of three commercially important target species of the Colombian Pacific coast, that contribute about one third to the total annual landings of small-scale fisheries in this coastal area (De la Hoz and Manjarrés-Martínez, 2016). Our approach is based on two set of indicators of stock condition that are well suited for data-poor fisheries since they require only length-frequency catch data, that is representative of the fisheries, and few external life history parameters that are generally available in the literature. The first set of indicators follows FAO's traditional stock assessment methods (Sparre and Venema, 1998) by estimating growth parameters, mortality rates and biological reference points but incorporates new procedures aimed at assessing the degree of uncertainty in the estimation of growth parameters using the recently developed R package TropFishR (Mildenberger et al., 2017; R-Core-Team, 2017). The second set of indicators uses catch proportions related to length-referenced points to assess the sustainability status of the fishery (Cope and Punt, 2009; Froese, 2004). We used official fisheries data collected by the national government (GOV) between 2013 and 2015 using a consistent sampling scheme carried out by the same institutional bodies (www.sepec.aunap.gov.co). Additionally, we carried out parallel stock assessments for the three species using data from a participatory fisheries monitoring program carried out by a non-government organization (NGO) in the northern sub-region of the Pacific coast from 2011 to 2013 (see Methods section). Our aim was to provide the best possible estimate of stock condition of the selected target species with the data currently available and explore differences among outcomes from different data sources that vary in the sampling scheme used and in their geographical extent.

2. Methods

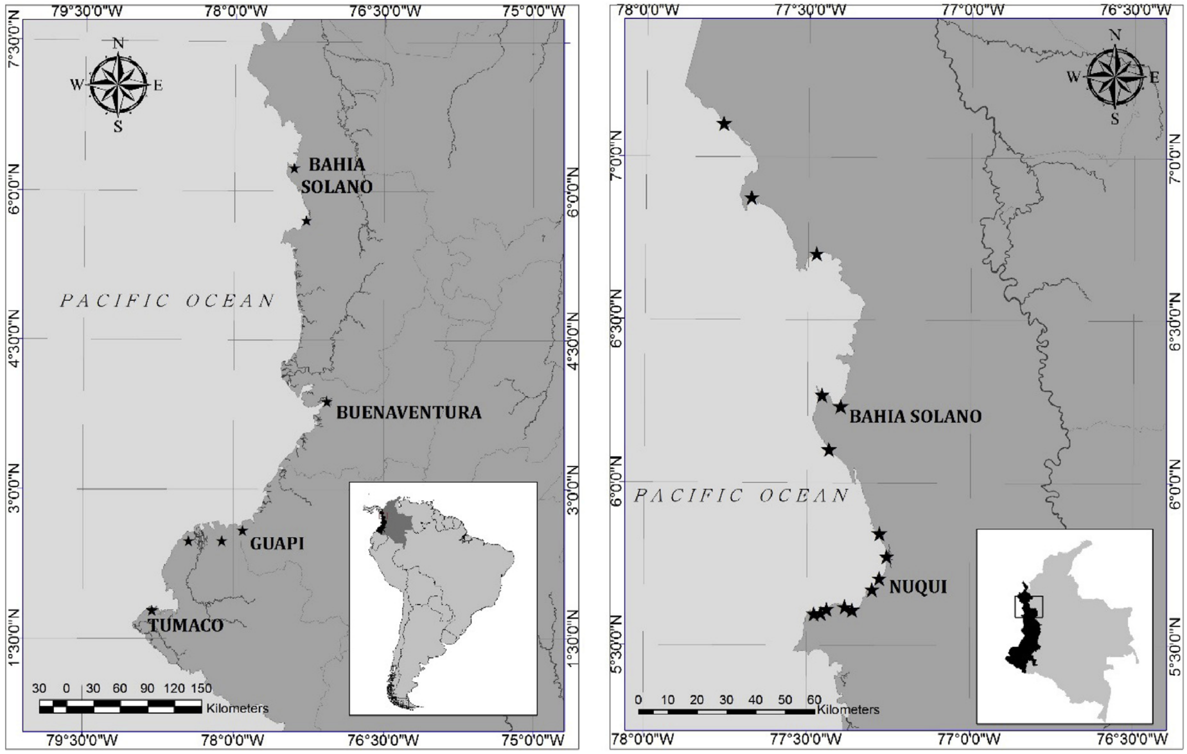
2.1. Study area

The Pacific coast of Colombia is located within the Panama Bight and belongs to the Eastern Tropical Pacific ecoregion. It stretches for ca. 1300 km (Correa and Morton, 2010) from the border with Panamá (7°13' 21"N, 77°53'25"W) to the border with Ecuador (1°27' 48"N, 78°51'43"W) (Fig. 1). This coastal region is characterized by high precipitation levels ranging 2500–9000 mm year^{-1}, abundant rivers that drain from the western Andes and a semi-diurnal tidal range, which varies between 3 and 4.5 m on spring tides and between 2 and 3 m on neap tides (IDEAM, 2005, 2014; Poveda et al., 2006). Sea surface temperatures in the region typically show two distinct periods, a colder one – associated to upwelling periods – from January to March, with temperatures averaging 26.5 °C, and a warmer part of the year with temperatures increasing up to 28.5 °C during the northern summer (Devis-Morales, 2009). However, events associated with El Niño Southern Oscillation can drastically change this seasonality (Wang and Fiedler, 2006). The northern Colombian Pacific sub-region extends for ca. 335 km from the border with Panamá (7°12'39" N, 77°53'21" W) to Cabo Corrientes in the south (5°29'50" N, 77°09'23" W). This coastal sub-region is distinguished from the rest of the Colombian Pacific by the predominance of rocky cliffs and sandy beaches and a relatively narrow continental shelf (1–15 km) which contrasts with the predominance of mangroves and estuaries, and average shelf width of 50 km in the central and southern sub-regions (Martínez et al., 1995; Castellanos-Galindo and Zapata, 2018). Additionally, the predominance of hook-based fishing gears in the northern sub-region contrasts with the predominance of gillnets in the rest of the Pacific coast, a situation that has been enhanced in recent years by community-driven management initiatives that aim to protect the rights of small-scale fishers (Díaz-Merlano et al., 2016; Ramírez-Luna and Chuenpagdee, 2019).

On the other hand, the Colombian Pacific region has a relatively low population density ranging from 5 to 17 people km^{-2} (Etter et al., 2006) and a low infrastructure development compared to the Caribbean coast of the country (Castellanos-Galindo and Zapata, 2018). There are only two large urban centers, one located in the center (Buenaventura) and the other one in the southern end of the coast (Tumaco), with several small rural villages scattered along the coastline and river basins (Fig. 1).

2.2. Selected target species

The stock assessment was carried out for the species Pacific sierra *Scomberomorus sierra* (Jordan and Starks, 1895), Spotted rose snapper *Lutjanus guttatus* (Steindachner, 1869) and Pacific bearded brotula *Brotula clarkae* (Hubbs, 1944), which together contribute to more than 30% of the total biomass landed by the small-scale fisheries in the Colombian Pacific (De la Hoz and Manjarrés-Martínez, 2016) . All three species are distributed along the entire range of the Eastern Tropical Pacific, i.e.: from the Gulf of California to the northern coast of Peru (Froese and Pauly, 2017), and have also been recognized as important species in the fisheries landings of other countries in the region such as: Costa Rica (Bystrom et al., 2017; Herrera et al., 2016; Naranjo-Elizondo et al., 2016), Panama (Vega et al., 2013) and Mexico (Amezcua et al., 2006; Espino-Barr et al., 2012). Despite their importance, very few assessments about the condition of their stocks have been published as peer-reviewed literature and, so far, none of them for the Colombian Pacific region (Amezcua et al., 2006; Espino-Barr et al., 2012). Table 1 summarizes biological and ecological features of the species as well as information related to their landings from small-scale fisheries in Colombia.



1A 1B

Fig. 1. Pacific coast of Colombia indicating the location of sampling sites included in the government (GOV) sampling scheme (1A), and sampling sites included by the non-government organization (NGO) sampling scheme (1B) during the study period. Names of main urban centers are included.

Table 1
Ecological characteristics of selected target species included in the present study, based on Froese and Pauly (2017), Robertson and Allen (2015) and Nielsen et al. (1999). Information on landings by the small-scale artisanal fisheries is also presented based on AUNAP and UNIMAGDALENA (2013a).

Species	Habitat	Maximum depth (m)	Key ecological features	Biomass landed (ton yr^{-1})	Main fishing gears	Percentage of total biomass landed
S. sierra	Pelagic–neritic	15	Swims in schools	634	Gillnets, handlines	17
L. guttatus	Demersal	120	Inhabits reefs (adults) and estuaries (juveniles)	272	Gillnets, handlines	7
B. clarkae	Bentho-pelagic	650	Inhabits soft-bottoms and rocky reefs	311	Longlines, handlines	8

2.3. Data collection

Length-frequency catch data (LFCD) from 2013 to 2015 was obtained from GOV data, based on sampling carried out at seven landing sites distributed along the Pacific coast (Fig. 1A). GOV sampling scheme included length measurements of random samples of fish at commercial landing sites, to the nearest 0.1 cm. Data from NGO was collected from 2011 to 2013 through a community-based fisheries monitoring program implemented by MarViva Foundation in the northern Pacific sub-region (Díaz-Merlano et al., 2016), including 13 rural and 2 urban landing sites (Fig. 1B). Total length of all fish arriving at each landing site was measured to the nearest 0.5 cm, two to three times per week. For both GOV and NGO, budget constraints, religious holidays or extreme weather conditions prevented data collection in some instances. Total number of sampling days per month and year in GOV and NGO data sets is included as part of the appendices (Table A1).

2.4. Stock assessment indicators

Assessment of the stock condition for the three selected species was carried out using two sets of indicators: (a) exploitation rate (E) and fishing mortality (F) relative to biological reference points, based on linearized catch curves and the yield per recruit (YPR) model, as described in Sparre and Venema (1998), and (b) length-based indicators of fishing sustainability, related to target points, as described in Froese (2004).

2.4.1. Growth parameters

Von Bertalanffy's growth parameters (VBGP) were estimated for each species, applying the seasonalised von Bertalanffy's growth function to the LFCD (Somers, 1988):

$$Lt = L\infty * (1\text{–}e\text{–}\ (K\ (t\text{–}t0) + S\ (t)\ \text{–}S\ (t0)))$$

where Lt is the length of the fish at a particular age t, $L\infty$ is the asymptotic length in cm, K is the growth rate coefficient in year^{-1}, t_0 is a theoretical age at which length is zero, $S(t) = (CK/2\pi) * sin2\pi(t - ts)$, C is a constant indicating the amplitude of the oscillation, typically ranging from 0 to 1, and ts is the fraction of a year (relative to the age of recruitment, $t = 0$) where the sine wave oscillation begins (i.e. turns positive) (Somers, 1988; Sparre and Venema, 1998). A bootstrapped version of the Electronic Length Frequency Analysis (ELEFAN) (Pauly and David, 1980, 1981; Pauly, 1982; Schwamborn et al., 2018) was used for the fitting process. In ELEFAN within TropFishR the parameter t_{anchor} is used to describe the fraction of the year where the von Bertalanffy growth function crosses length = 0 for a given cohort (Taylor and Mildenberger, 2017). Additionally, the growth performance index (Φ') was also

Table 2
Total number of fish measured by the government (GOV) sampling scheme from 2011 to 2013 and by the non-government organization (NGO) sampling scheme from 2013 to 2015. Size range, mean length and percentage (%) of fish landed, according to the type of fishing gear used, are also presented.

Species	Data source	Total *n*	Size range (cm)	Mean length (cm)	% per gear type			
					Gillnet	Hand line	Longline	Others
S. sierra	GOV	16531	11–110	46.2	89.6	8	0.2	2.2
	NGO	20646	14–99.5	53.2	70.2	29.3	0.3	0.2
L. guttatus	GOV	16952	9–99	35.9	55.3	34.7	8.0	2.1
	NGO	58026	6.5–89.5	35.1	55.6	43	1.4	0
B. clarkae	GOV	4273	6.5–100	71.7	0.4	52.5	47.2	0
	NGO	18574	17.5–130	73.4	0.0	2.7	97.3	0

Table 3
Estimated values for VBGP (maximum density—max. dens.) and confidence intervals (upper interval and lower interval) resulting from bootstrapped ELEFAN analysis for the three selected target species, based on government (GOV) and non-government (NGO) data sources.

Species	Parameters	GOV			NGO		
		Max. dens.	Lower int.	Upper int.	Max. dens.	Lower int.	Upper int.
S. sierra	$L\infty$	88.97	81.92	113.60	74.89	72.07	105.24
	K	0.41	0.08	0.56	0.22	0.06	0.53
	t_{anchor}	0.45	0.03	0.91	0.53	0.06	0.92
	C	0.34	0.09	0.98	0.34	0.09	1.00
	ts	0.66	0.02	0.99	0.63	0.00	0.95
	Φ'	3.51	2.73	3.86	3.09	2.49	3.77
L. guttatus	$L\infty$	59.61	54.73	77.63	65.85	50.12	73.73
	K	0.57	0.34	1.22	0.47	0.14	1.78
	t_{anchor}	0.44	0.15	0.92	0.47	0.01	0.97
	C	0.49	0.05	0.99	0.63	0.18	1.00
	ts	0.52	0.04	1.00	0.72	0.01	0.98
	Φ'	3.31	3.01	3.87	3.31	2.55	3.99
B. clarkae	$L\infty$	90.65	83.25	119.78	88.32	86.25	91.39
	K	0.23	0.07	0.97	0.25	0.14	0.41
	t_{anchor}	0.50	0.10	0.94	0.59	0.03	0.88
	C	0.57	0.01	0.99	0.63	0.12	1.00
	ts	0.38	0.02	1.00	0.24	0.01	1.00
	Φ'	3.28	2.69	4.14	3.29	3.01	3.54

estimated as: $\Phi' = \log(K) + 2 * \log(L\infty)$, based on (Pauly, 1984). An initial seed value of $L\infty$was estimated based on *Lmax*, derived from the mean of the 1% largest fish in the sample and following the formula from Pauly (1984): $L\infty = Lmax/0.95$.

In order to improve cohort visualization, LFCD was filtered to 14-day periods within each month based on target days that were selected for each species and data source through an application developed using the shiny R-package (https://shiny.rstudio.com) (Chang et al., 2017). The use of this application allowed us to readily quantify and visualize the sample size per day of the month (1 to 31), for each combination of species, year and data source. A table summarizing the target days selected for each species and data source is included in the appendices (Table A2). The most suitable moving average (MA) value for each data source was determined by restructuring the data based on different MA values and the rule of thumb established in Taylor and Mildenberger (2017) concerning the number of bins spanning the youngest cohorts. The initial seed value $L\infty \pm 20\%$, and a K range between 0.01 and 2 defined the search space for the 500 resamples of the bootstrapped ELEFAN. For all additional parameters (t_{anchor}, C, t_s), which are bound between 0 and 1, the search space was not further limited but spanned the whole unit interval. Maximum density estimates and 95% confidence intervals for all VBGP were obtained for each species and data set, based on the 500 resamples of the bootstrapping procedure.

2.4.2. Exploitation rate and fishing mortality

Once VBGP were estimated, the entire data (i.e.: not filtered) was used for subsequent analysis. In order to account for missing data and make LFCD more representative of real catches, raising factors, as defined in Sparre and Venema (1998), were estimated for each species and data source based on: (a) days not-sampled in any given month, taking into account monthly variations in the catch of each species, (b) catch used for local consumption, based on the estimates made by Wielgus et al. (2010), and (c) proportion of fishing trips sampled versus total fishing trips per day based on data recorded by MarViva Foundation at each landing sites. A table summarizing the end value and the method to calculate raising factors used for each species and data set source presented in the appendices (Table A3).

Using raised LFCD and previously estimated VBGP, linearized length-converted catch curves were produced for each species and data source taking into account growth seasonality. Catch curves were applied to the average catch numbers per length class across all years, and for the year 2013 only, being this the only common year between the data sources. Total instantaneous mortality rate (Z) was estimated by calculating the slope of the regression line of the descending part of the catch curve. The selection of points for the regression line was based on the age (length-derived) classes represented in the catch in each case (i.e.: species/data-source combination).

The rate of natural mortality (M) was estimated using the empirical formula developed by Then et al. (2015): $M = 4.118K^{0.73}L_{\infty}^{-0.33}$

This formula was preferred over the wide range of available empirical formulas for estimating M, since it yielded better prediction power for more than 200 species of fish with different life histories, when accurate estimations of maximum age are not readily available. Based on the estimated Z and M values, fishing mortality (F) and exploitation rate (E) were estimated from: $F = Z - M$ and $E = F/Z$, respectively. Estimated values of E were then compared to a reference value of 0.5, which has been proposed as an upper level of sustainable exploitation for fish species (Gulland, 1971). Estimated F values were also compared against reference

Table 4
Estimated values of natural mortality (M), fishing mortality (F), biological reference points of fishing mortality (F_{max}, $F_{0.1}$, $F_{0.5}$) and exploitation rate (E) for the selected target species, based on government (GOV) and non-government (NGO) data sources. Estimations were carried out for the average catch of three years within each data source and for 2013 only.

Species	Parameters	GOV		NGO	
		2013–2015	2013	2011–2013	2013
	M	0.49	0.49	0.49	0.49
	Z	1.04	1.06	1.43	1.41
	F	0.55	0.57	0.95	0.92
S. sierra	E	0.53	0.54	0.66	0.65
	$F_{0.1}$	0.29	0.29	1.01	1.04
	F_{max}	0.54	0.55	1.01	1.35
	$F_{0.5}$	0.20	0.20	0.17	0.17
	M	0.71	0.71	0.71	0.71
	Z	1.01	1.05	1.78	1.76
	F	0.30	0.34	1.07	1.05
L. guttatus	E	0.30	0.32	0.60	0.60
	$F_{0.1}$	0.62	0.74	0.70	0.91
	F_{max}	0.87	1.02	0.97	1.23
	$F_{0.5}$	0.29	0.29	0.29	0.29
	M	0.34	0.34	0.34	0.34
	Z	0.23	0.26	0.29	0.28
	F	−0.11	−0.08	−0.05	−0.06
B. clarkae	E				
	$F_{0.1}$	0.20	0.22	0.20	0.20
	F_{max}	0.63	0.67	0.59	0.61
	$F_{0.5}$	0.14	0.14	0.14	0.14

points derived from the YPR prediction model of Beverton and Holt (1957): (a) the highest biomass per recruit (F_{max}), (b) a 50% reduction of the biomass of unexploited population ($F_{0.5}$), and (c) a fishing mortality which corresponds to 10% of the slope of the yield per recruit curve in the origin ($F_{0.1}$). Parameters of the length–weight relationship, *a* and *b,* required as inputs for the YPR model, were obtained from recent estimations carried out by the Colombian fisheries authority (Puentes et al., 2014) that can be found in FishBase (Froese and Pauly, 2017).

2.4.3. Length-based indicators for fishing sustainability

The second set of indicators for the assessment of stock condition are the simple three length-based indicators (LBI) for fisheries sustainability and their associated reference points synthesized as "let them spawn, let them grow and let the mega-spawners live" (Froese, 2004). These LBI are based on previous studies that have documented links between the variables involved in the estimation of the LBI with recruitment overfishing and/or growth overfishing (Barneche et al., 2018; Berkeley et al., 2004; Beverton, 1992; Myers and Mertz, 1998). The LBI used here are:

(a) *Pmat*, the proportion of mature fish in the catch, with 100% as the reference target point, based on the formula: $Pmat = \%$ fish in sample $>Lm$; where Lm is the length at maturity. Lm values used here were derived from a relatively recent assessment carried out by Colombia's fisheries authority, where they estimated median length at maturity (estimated length at which 50% of the fish are mature) for several target species in the country, based on visual assessment of gonads stage of maturity and the use of a logistic model to assess proportion of mature fish per length class, based on data collected in 2013 (Puentes et al., 2014). Total length Lm values used here were: 58.8 cm for *S. sierra*; 35.3 cm for *L. guttatus* and 75.4 cm for *B. clarkae*.

(b) *Popt*, the proportion of fish within a 10% range around the optimum length ($Lopt$) in the catch, with 100% as the reference target, based on the formula: $Popt = \%$ fish $\geq Lopt - 10\%$ and $\leq Lopt + 10\%$; where: $\log(Lopt) = 1.053{*}\log(Lm) - 0.0565$, based on (Froese and Binohlan, 2000). Estimated $Lopt$ for the selected target species, based on this formula, were: 63.6 cm for *S. sierra*; 37.2 cm for *L. guttatus* and 82.7 cm for *B. clarkae*.

(c) *Pmega*, the proportion of "mega-spawners" in the catch, with 30%–40% considered acceptable percentages in the catch when no specific management strategy is in place, based on the formula: $Pmega = \%$ fish $> Lopt + 10\%$ Froese (2004).

The three proportions, or LBI, were then summed ($Pmat + Popt + Pmega$) to obtain *Pobj*, a combined indicator used to follow a decision-tree designed by Cope and Punt (2009), which could prove useful in multi-gear fisheries where the assumption of trawl-like selectivity is often not fulfilled. This decision tree is based on the results of a deterministic population dynamics model developed by the authors to explore the effects of different fishery selectivity patterns, different recruitment compensation rates and different life history traits, on the outputs of the LBI proposed by Froese (2004). Through their model, the authors found that *Pobj* had a more consistent relationship with spawning biomass (SB) than any of the individual LBI (*Pmat*, *Popt* or *Pmega*), and that different selectivity patterns in the fishery were associated to range of values of *Pobj*. Once a selectivity pattern is established based on *Pobj*, threshold values of *Pmat, Pobj* and/or the $Lopt/Lm$ ratio point to an estimated probability of the stock spawning biomass (SB) being below established reference points, either 40% or 20% of the unfished spawning biomass (0.4SB or 0.2SB). For further details please refer to Fig. 10 and Table 5 in Cope and Punt (2009).

3. Results

A total of 135,002 fish were included in the analysis: 37,177 for *S. sierra*, 74,978 for *L. guttatus* and 22,847 for *B. clarkae*. Gillnets of varied mesh sizes (2.5–20 cm), hand lines and longlines with hooks of different sizes (#5 to #9), were the predominant gears employed in the fisheries, although their proportions differed among the two data sources (Table 2). As described above, hand lines and longlines are more commonly used in the northern sub-region of the Colombian Pacific (NGO data) than in the rest of the coast.

3.1. Growth parameters

Estimated VBGP and confidence intervals (CI) were obtained through the bootstrapped ELEFAN based on 500 resamples (Fig. 2, Table 3). An MA of 9 was selected for all three species. Growth curves obtained through ELEFAN using two additional MA values are included as appendices (Fig. A1). Estimated confidence intervals cover a wide range indicating an overall high uncertainty in the estimations. Particularly, upper CI bounds show a larger deviation from the estimated values (maximum density result after 500 resamples) than lower bounds. However, results based on GOV data showed narrower confidence intervals than those from NGO for *S. sierra* and *L. guttatus*, whereas NGO results had narrower intervals for *B. clarkae* (Table 3). Differences are also observed between the VBGP estimated from GOV and NGO data, with largest differences found for *S. sierra*. For this species GOV data resulted in substantially higher values of $L\infty$, K and Φ'. For *L. guttatus* and *B. clarkae* Φ' was similar between data sources, while $L\infty$, K values also differed.

Estimated C values were similar between the two data sources for the three species (Table 3), which suggests seasonal oscillations in growth rate for all of them. Estimated t_s values ranged from 0.5 to 0.7 for *S. sierra* and *L. guttatus* and between 0.2 and 0.4 for *B. clarkae* (Table 3). These periods correspond to June–August and February–May, respectively. It must be noted that parameters t_{anchor} and t_s are defined in the unit interval (i.e.: range from 0 to 1) and due to the yearly repeating pattern of the growth curves, a growth curve

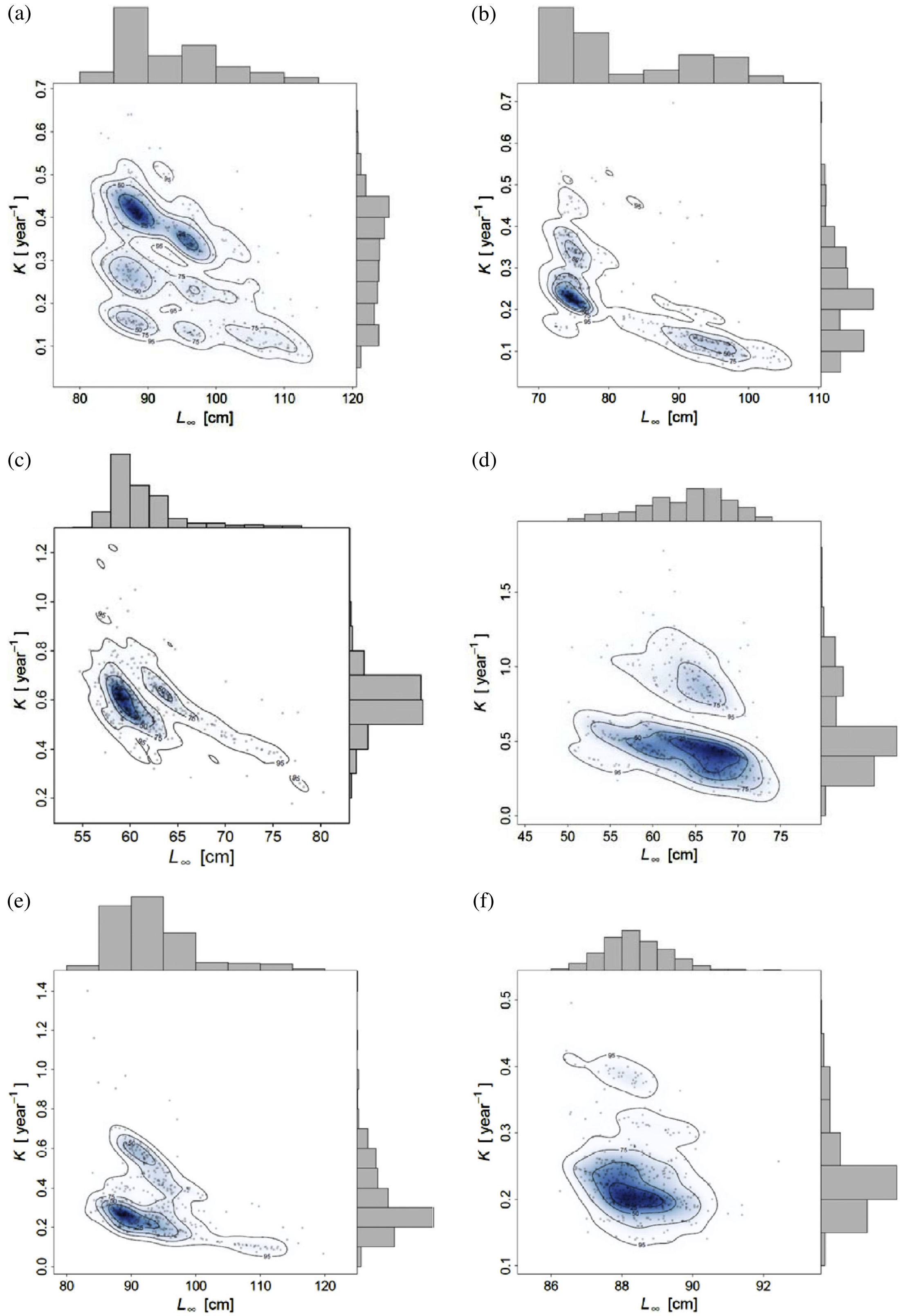


Fig. 2. Graphic outcome of bootstrapped ELEFAN for the three selected species and the two data sources (government data — GOV) and non-government data — NGO), using TropFishR. Dots represent estimated $L\infty$ and K (growth parameters of the von Bertalanffy's equation) per resampled length-frequency catch data. Marginal histograms show univariate density for both parameters, while density lines and color intensity indicate the multivariate density. (a) *S. sierra* — GOV, (b) *S. sierra* — NGO, (c) *L. guttatus* — GOV, (d) *L. guttatus* — NGO, (e) *B. clarkae* — GOV, (f) *B. clarkae* — NGO.

with a t_{anchor} or t_s value of 0.01 is similar to one with a value of 0.99. This is reflected in the large confidence intervals that resulted in the estimation of those two parameters and of parameter C, which is strongly dependent on t_s.

3.2. Fishing mortality and exploitation rate

Only one set of the previously estimated VBGP – either GOV or NGO derived – was selected as input for the linearized length-converted catch curve and YPR analyses for each species. Preference was given to: smaller variation in outputs across different moving averages, narrower range of confidence intervals and larger sample size. Therefore, VBGP estimated values derived from GOV were used for *S. sierra* and *L. guttatus*, whereas VBGP estimated values derived from NGO were used for *B. clarkae*. Table 4 summarizes the estimated values of *M, Z, F, E* and biological reference levels of fishing mortality (F_{max}, $F_{0.1}$ and $F_{0.5}$) estimated for each species, data source and time period. Graphical outputs of all catch curves produced are included in the appendices (Fig. A2 and A3).

The estimated *M* value was constant for each species since only one set of estimated VBGP was selected and used for its calculation from the selected empirical formula (Then et al., 2015). *Z* values were very similar between time periods (i.e.: average catch of three years versus catch of 2013) within each combination of species/data source, which suggests little or no variation in *F* among years. A higher *E* was observed in the values estimated from NGO for *S. sierra* and *L. guttatus*, compared to those estimated from GOV. In the case of *S. sierra*, estimated values of *E* derived from both data sources are above the threshold of $E = 0.5$, used as an indicator of over-exploitation (Gulland, 1971). In contrast, estimated values of *E* for *L. guttatus* indicate an over-exploitation status based on NGO, but an under-exploited status based on GOV (Table 4).

Comparisons of estimated *F* values with regard to biological reference points show a similar diagnosis of stock condition to *E* results for *L. guttatus* but not so for *S. sierra* (Table 4). Estimated *F* for *S. sierra* derived from GOV is above the biological reference points of fishing mortality F_{max} and $F_{0.1}$, suggesting over-exploitation. However, NGO derived results for this species, indicate that *F* is close to, but still lower than, the F_{max} and $F_{0.1}$estimated through YPR analysis, which would indicate a fully exploited, but not yet over-exploited, stock. Unexpectedly, estimated *Z* for *B. clarkae* based on both data sources had a lower or similar value than *M*, resulting in negative *F* values close to 0. This is not realistic and believed to be an artifact of the methods used and quality and quantity of the data (see Discussion). Therefore, no real diagnosis can be made at this point about the stock condition of this species based on the first set of indicators and the data available, other than the fishing mortality might be relatively low.

3.3. Length-based indicators of fishing sustainability

Table 5 summarizes the estimated *Pmat*, *Popt* and *Pmega* in the catch for each species and data source. None of the estimated proportions comply with the individual reference target values proposed by Froese (2004) for this indicators. Nevertheless, results derived from NGO data might be interpreted as suggesting a better condition of the stocks than results from GOV, since *Pmat* and *Popt* values were higher in the former than in the latter. Specifically, *L. guttatus* and *B. clarkae* show more than 50% of mature and optimum-sized fish based on NGO data. On the other hand, *Pmega* values are very low for *S. sierra* and *B. clarkae*, which could be interpreted as desired values in a healthy fisheries that is "letting" the older or mega-spawners out of the catch (Barneche et al., 2018; Mora et al., 2009; Myers and Mertz, 1998). However, after applying the decision-tree proposed by Cope and Punt (2009), which accounts for the intrinsic selectivity pattern of the fisheries, results indicate that stocks of *L. guttatus* and *B. clarkae* have spawning biomass below target reference point of 0.4 SB with a probability of 100%. In the case of *S. sierra*, results are quite ambiguous since those derived from NGO indicate that the stock is in good condition but results from GOV indicate that there is a 52% probability that the spawning biomass is below the target reference point of 0.4 SB.

4. Discussion

In many tropical developing countries small-scale fisheries do not play a significant role in the national economic performance (e.g.: GDP — Gross Domestic Product) – despite their high importance in terms of local income, employment and food security for coastal communities – and therefore government allocation of resources for fisheries management is very scarce (Kato et al., 2012; Purcell and Pomeroy, 2015). This often results in a very limited quantity and quality of data available for adequate assessments of the status of fisheries resources, and managers are faced with the challenge of making decisions, with very limited information, for nearly 99% of the species reported in the worldwide catch (Costello et al., 2012; Honey et al., 2010). In this study, we attempted to provide the best assessment possible for three target species of the small-scale fisheries in the Eastern Tropical Pacific, using LFCD collected in Colombia, a country with all of the fisheries management constraints and the data limitations described above (Castellanos-Galindo and Zapata, 2018; Saavedra-Diaz, 2012). We did so, combining two methodological approaches suited for data-poor contexts, taking into account that a single biological reference point is not sufficient in providing an unequivocal diagnosis of stock condition, particularly when dealing with several sources of uncertainty (Erisman et al., 2014; Ramírez et al., 2017). In the following paragraphs, we discuss the outcomes of the assessment routines and their implications for fisheries management.

4.1. Growth parameters

Our estimation of growth parameters based on von Bertalanffy's growth equation followed traditional methods of length frequency stock assessments (Sparre and Venema, 1998) but incorporated novel steps to improve the precision of the estimates and to measure the degree of uncertainty. The new procedures comprised: (1) an initial "seed" $L\infty$ input value estimated using the 1% largest fish in the catch, with the aim of reducing the effect of outliers in the sample; (2) an active selection of the length group's moving average value to enhance cohorts' visualization in the reconstructed length-frequency plots, and (3) the application of ELEFAN with improved optimization routines within a bootstrapping framework that allows to estimate confidence intervals of growth parameters estimations (Mildenberger et al., 2017; Schwamborn et al., 2018; Taylor and Mildenberger, 2017).

Our estimated values of $L\infty$, *K* and Φ' fall within the range of previous estimations made for the three target species (Table 6), with two exceptions. First, the estimated asymptotic length for *B. clarkae* was lower than previously estimated in Colombia by the national fisheries authority, even though the search range input used for the bootstrapped ELEFAN analysis (81–121 cm) was wide and covered most previous literature estimates (Table 6). The second exception was the estimated *K* value for *L. guttatus* which was higher than previously estimated from fisheries targeting this species in the Eastern Tropical Pacific. In the case of *B. clarkae*, fishes are predominantly caught by hook-based gears (hand lines and longlines) that impose a specific type of selectivity reflected in the size distribution of the catch, which influence the visualization of cohort progression in the ELEFAN analysis (Fig. 3, Fig. A1). Previous studies carried out for this species in Colombia included catches from industrial shrimp-trawling fisheries, which might have resulted in different size distributions and therefore different estimations of VBGP (Angulo, 1995; Angulo and Zapata, 1997).

On the other hand, the relatively high value estimated for *K* in *L. guttatus* seems to compensate for a relatively low $L\infty$ estimated for this species, which is below the maximum reported length for this species (80 cm) (Froese and Pauly, 2017). Despite the

Table 5
Proportions of mature fish (*Pmat*), optimum-sized fish (*Popt*), larger than optimum size fish (*Pmega*) and *Pobj* (= *Pmat* + *Popt* + *Pmega*) for each species and data source (government — GOV or non-government organization — NGO), based on the indicators proposed by Froese (2004) and the formulas described in Methods. Stock condition interpretation is based on a decision tree proposed by Cope and Punt (2009), aimed to assess whether spawning biomass (SB) is above (>) or below (<) a reference point (RP) of 0.4 unfished biomass. The last column indicates the estimated probability of SB being lower than 0.4 of unfished biomass based on the same authors.

Species	Data source	*Pmat*	*Popt*	*Pmega*	*Pobj*	Stock condition interpretation	Probability
S. sierra	GOV	0.15	0.11	0.06	0.32	SB < RP	52%
	NGO	0.38	0.38	0.06	0.82	SB ≥ RP	0%
L. guttatus	GOV	0.47	0.28	0.27	1.02	SB < RP	100%
	NGO	0.56	0.52	0.15	1.23	SB < RP	100%
B. clarkae	GOV	0.48	0.44	0.05	0.97	SB < RP	100%
	NGO	0.51	0.53	0.01	1.05	SB < RP	100%

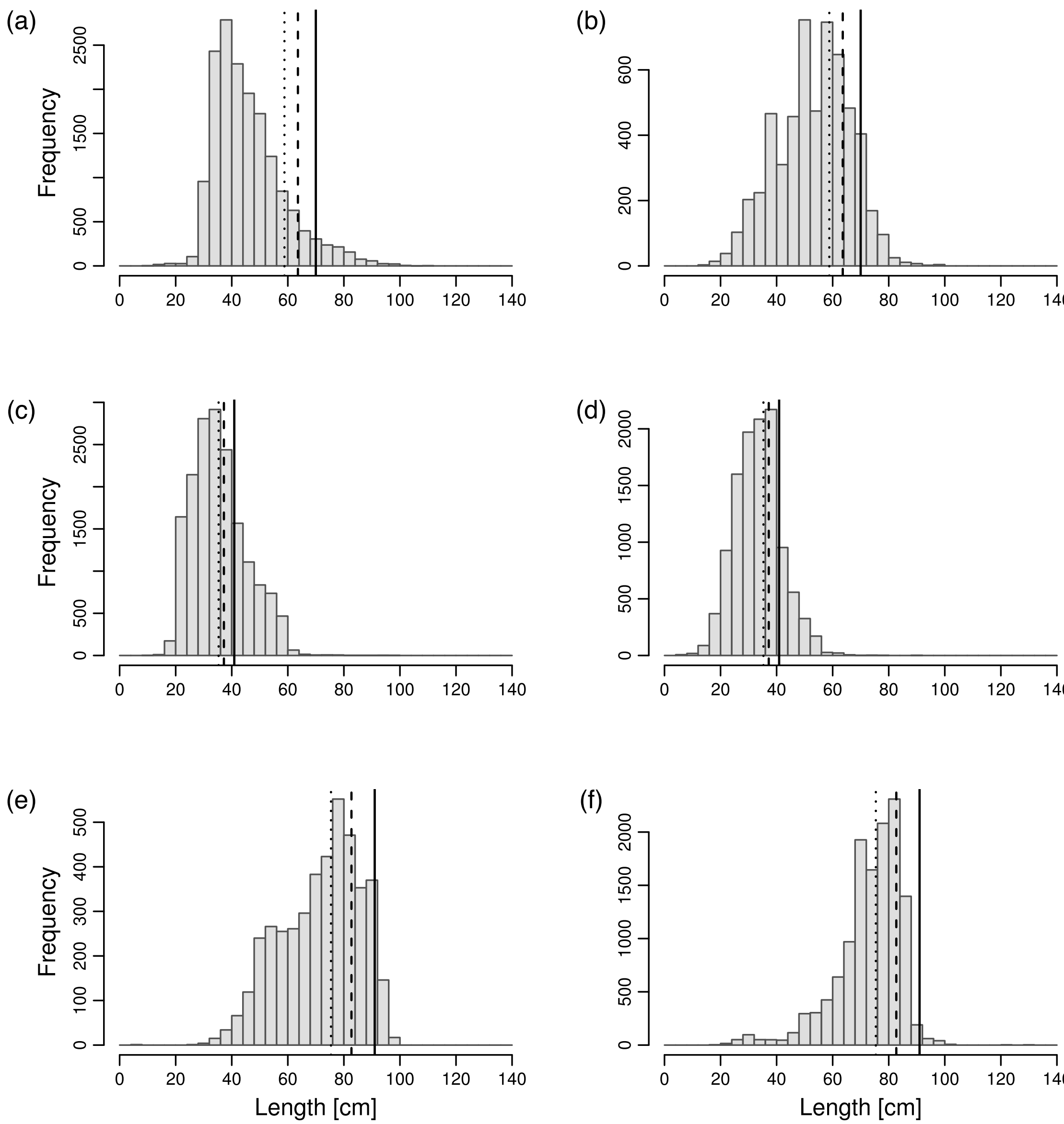


Fig. 3. Size distribution of landed fish of the three target species assessed here based on two data sources: government data (GOV) and data from a non-government organization (NGO). Vertical lines indicate length-based reference values of: length at maturity *Lm* (dotted line), optimum length *Lopt* (dashed line) and 10% above the optimum length *Lopt*+10%, (solid line). (a) *S. sierra* — GOV, (b) *S. sierra* — NGO, (c) *L. guttatus* — GOV, (d) *L. guttatus* — NGO, (e) *B. clarkae* — GOV, (f) *B. clarkae* — NGO.

existing records of large individuals of *L. guttatus* in the wider Eastern Pacific region that are close, or even >80 cm, several studies carried out on this species, that were based on catch data, show maximum lengths of 50–60 cm (Andrade-Rodriguez, 2003; Barreto and Borda, 2008; Sarabia-Méndez et al., 2010; Soto Rojas et al., 2009), which suggests that either the selectivity imposed by the fisheries exclude the largest size classes or that those large individuals (>60 cm) are now rare cases or "outliers" in the natural population, due to historical fishing impact. Our estimated $L\infty$value is similar to those estimated using otolith-derived age-frequency data from catch data from the Mexican Pacific (Amezcua et al., 2006; Andrade-Rodriguez, 2003; Soto Rojas et al., 2009), that could also support the "outliers" hypothesis.

Table 6
Comparison of estimated values of growth parameters $L\infty$, K and Φ' for the three target species assessed here. All studies were carried using length-frequency catch data, except those indicated (†) that were based on reading of otoliths, or other hard structures, to estimate age.

Species	$L\infty$	K	Φ'	Source	Country
S. sierra	88.97	0.41	3.51	Present study (GOV)	Colombia
	74.89	0.22	3.09	Present study (NGO)	Colombia
	85.81	0.37	3.44	Puentes et al. (2014)	Colombia
	108.3	0.15	3.25	Aguirre-Villaseñor et al. (2006)	Mexico
	111.3	0.16	3.30	Barreto et al. (2010)	Colombia
	111.2	0.16	3.30	Barreto and Borda (2008)	Colombia
	75.10	0.19	3.03	Medina Gomez (2006)†	Mexico
	99.54	0.21	3.31	Nava-Ortega et al. (2012)†	Mexico
	81.00	0.6	3.60	Castillo (1998)	Colombia
L. guttatus	59.61	0.57	3.31	Present study (GOV)	Colombia
	65.85	0.47	3.31	Present study (NGO)	Colombia
	79.50	0.4	3.40	Puentes et al. (2014)	Colombia
	66.19	0.13	2.76	Amezcua et al. (2006)†	Mexico
	139.00	0.28	3.73	Barreto et al. (2010)	Colombia
	87.5	0.24	3.26	Barreto and Borda (2008)	Colombia
	64.58	0.21	2.94	Bystrom et al. (2017)	Costa Rica
	66.40	0.13	2.76	Andrade-Rodriguez (2003)†	Guatemala
	96.60	0.26	3.38	Sarabia-Méndez et al. (2010)	Mexico
	65.90	0.13	2.75	Soto Rojas et al. (2009)†	Costa Rica
	55.10	0.40	3.08	Suárez (1992)	Colombia
B. clarkae	90.65	0.23	3.28	Present study (GOV)	Colombia
	88.32	0.25	3.29	Present study (NGO)	Colombia
	103.80	0.50	3.73	Puentes et al. (2014)	Colombia
	119.20	0.20	3.45	Barreto et al. (2010)	Colombia
	118.00	0.70	3.99	Angulo (1995)	Colombia
	130.00	0.50	3.93	Angulo and Zapata (1997)	Colombia
	105.20	0.45	3.70	Muñoz (1999)	Colombia

Our results show very large confidence intervals in estimated growth parameters, particularly in the estimation of *K*, the growth coefficient. A wide range of estimated values of $L\infty$ and *K* parameters have also been observed in previous studies carried out for these species in Colombia and in other countries of the Eastern Tropical Pacific (Table 6). These studies have shown that estimations of length-derived VBGP can be highly influenced by the bias imposed by the selectivity of fisheries (either gear imposed, spatial or temporally driven), since the catches do not necessarily represent the size structure of the natural populations (Maunder et al., 2014; Punt et al., 2014; Sampson, 2014; Taylor and Martell, 2005). Using otolith-derived size at age estimates, Medina Gomez (2006) found different values of growth parameters of *S. sierra* from different coastal zones within the Gulf of California in the Pacific coast of Mexico and related these findings to the selectivity of fishing gears used in the different zones, but not excluding the potential existence of sub-populations along the coastal region. Different stocks of the same species had been identified within a single coastal region, such as the case of the congeneric *S. cavalla* in the Gulf of Mexico (Johnson et al., 1994). In our case, the northern sub-region of the Colombian Pacific presents particular environmental features compared to the rest of the coast (see Methods), and there are also differences in the proportion of fishing gears reported in the landings of the two geographical areas included in the present study (Table 2). Therefore, both factors – gear imposed differences in selectivity between zones and spatial differences of size distribution among subpopulations – could contribute to the different patterns of size structure and related cohort's progression observed in the two data sources (Fig. 3, Fig. A1).

Another difference between our VBGP estimation method and others previously carried out for these species, is the fact that we used the seasonalised growth function for the ELEFAN analysis. While the main VBGF parameters ($L\infty$, K) and associated confidence intervals presented here are not substantially different from those derived from the non-seasonal approach (Table A4), our results suggest that there are seasonal changes in the growth rates of the three species, which could be related to changes in water temperature, precipitation and/or to the availability of food in the coastal shelf zone (Morales-Nin and Panfili, 2005). In the case of *S. sierra* and *L. guttatus*, results indicate that the period with the highest growth rate falls between the months of June and August, which could be due to the period of warmer sea-surface temperatures in the Colombian Pacific (Devis-Morales, 2009). In the case of *B. clarkae*, a species that inhabits deeper water habitats, different environmental factors could be influencing the growth rate during the first months of the year (Feb–Mar), but these factors are currently unknown.

4.2. Stock condition based on estimates of E and F

Confidence intervals for VBGP are not yet routinely incorporated into length-based stock assessments and it is probable that the high uncertainty in the VBGP found in our study may also be underlying many past length-based stock assessments. This quantified uncertainty provides helpful guidance on the interpretation of the result of subsequent analyses; due to the wide confidence intervals around the VBGP, resulting estimates of *E*, *F* and biological reference levels should be regarded as a first approximation to the current stock condition of the target species assessed here.

Our results suggest that different sampling schemes could lead to different outcomes of stock condition and may end up giving contrasting guidelines to managers. Estimated values of *E* and F/F_{max} for *S. sierra* and *L. guttatus* differed somewhat between the two data sources used. For *S. sierra*, results derived from both data sets suggest a status of overexploitation, although NGO estimates were higher. Also in *L. guttatus*, NGO data provided higher estimates for *E* and *F* but in this case those data suggested overexploitation, while the far lower estimates derived from the GOV data rather indicated under-exploitation (Table 4). In the case of *B. clarkae*, estimated *Z* was very close to or even lower than *M*, which could imply that fishing mortality is almost negligible for this species and that the existing biomass is under-exploited. However, from the catch curves (Fig. A2 and A3) it seems that the regression line and thus *Z* does not represent well the entire data, since it might be overestimating mortality for smaller individuals and underestimating it for larger ones. This unusual pattern of the catch curve can most probably be attributed to different gears (and/or mesh and hook sizes) and associated selectivities (Table 2), but could also have resulted from differences in the availability of different cohorts at a particular time or place. Since the catch curve method is based on the assumption that beyond a critical size, all larger fish are fully retained, catch-at-length matrices should ideally be corrected before running the catch curve analysis, based on the knowledge of the selectivity curves of each fishing fleet involved (Punt et al., 2014; Sampson, 2014). Unfortunately, the lack of knowledge on the selectivity features of the different fleets involved in the fisheries, and the unavailability of fisheries-independent data on natural abundance and/or reproductive processes did not allow us to correct catch-at-length data in the present study.

Stock assessments carried out by Colombia's fisheries authority from 2008 to 2016, based on catch curves and surplus production models, reported an overexploitation status for the three target species assessed here (AUNAP, 2016; Barreto and Borda, 2008; Barreto et al., 2010; Puentes et al., 2014). The most comprehensive of those assessments – based on data collected in 2013 during the entire year – reported *E* values of 0.59, 0.72 and 0.76 for *S. sierra*, *L. guttatus* and *B. clarkae,* respectively (Puentes et al., 2014). Official total landings data for each species from 2006 to 2017 do not show a clear trend of relative stock abundance (Fig. A4), but it must be noted that these data are not readily comparable because of the lack of information on fishing effort and the lack of consistency in sampling frequency and spatial stratification (www.sepec.gov.co). As described above, these inconsistencies are mainly linked to budget constraints and drastic institutional changes that occurred within the fisheries government sector of the country between 2002 and 2011.

Besides the technical reports produced by the Colombian fisheries authority, very few stock assessments have been published in the literature for the three target species. Even though the stocks of neighboring countries within the Eastern Tropical Pacific could be different from the ones exploited in the Colombian Pacific, their assessments could provide another reference. In the central Pacific of Mexico, Espino-Barr et al. (2012) found an exploitation rate of 0.74 for *S. sierra* based on catch data, with a higher F than F_{max} (0.57 vs 0.42), indicating a status of overexploitation of this species in this country. On the other hand, for *L. guttatus*, Amezcua et al. (2006) reported an E between 0.19 and 0.43 for the southern Gulf of California in Mexico and Bystrom et al. (2017) reported an exploitation rate of 0.44 based on samples taken from the northern coast of Costa Rica. Both studies found exploitation rates slightly below the threshold of $E = 0.5$, which resembles the output of our assessment for the Colombian Pacific based on GOV data.

4.3. Stock condition based on LBI

Even though the final stock diagnosis based on LBI showed more consistent results among the two data sources than the outputs of E and F, there were also slight variations between the estimates derived from the two data sources, which further suggests that different sampling schemes may lead to different management advice. Particularly, in the case of *S. sierra* the diagnosis was not consistent, since results from NGO data gave a picture of a healthy stock, while GOV data indicate a 52% probability that the spawning biomass is below the reference target level. A key difference observed among the two data sources for *S. sierra* was the higher proportion of mature fish (*Pmat*) in NGO data compared to GOV data (Table 5). This difference in *Pmat* results in two different diagnosis of stock condition when applying the decision tree that, in this case, requires a $Pmat > 0.25$ to conclude that the spawning biomass is above the target level (Cope and Punt, 2009). In contrast, for both *L. guttatus* and *B. clarkae* the combined indicator *Pobj* and the final diagnosis indicate that there is a 100% probability that the spawning biomass is below the reference target level (0.4SB).

All LBI used here as a second set of indicators to assess the stock condition of the target species were based on *Lm* values previously estimated in the country (see Methods) and therefore were not influenced by our estimated VBGP. The estimation of the combined indicator *Pobj* and the application of the decision tree – as proposed by Cope and Punt (2009) – have not been previously carried out before for the three selected target species. Nonetheless, it is possible to compare our estimated proportions of mature individuals (*Pmat*) in the catch with previous estimates made in Colombia and in other countries of the Eastern Tropical Pacific. This comparison, though, must consider the existing variations in the estimated *Lm* used as reference value in different studies (Table 7). It has been shown that differences in *Lm* within the same species can be related to environmental or latitudinal factors, genotype, stock size and/or historic fishing pressure (Cardinale and Modin, 1999; Heibo et al., 2005; Rowell, 1993). Differences observed in *Pmat* estimates are influenced not only by their reference *Lm* value but also by the type of fisheries involved in the sampling. For example, estimated *Pmat* for *B. clarkae* derived from industrial trawling shrimp fisheries in Costa Rica (Herrera et al., 2016) is much lower than *Pmat* values estimated in Colombia where most catches come from longlines and hand lines. Thus, gear selectivity, habitats used as fishing grounds and environmental factors that could trigger ontogenetic movements impose a bias in the size distribution of the catch and in turn a bias on the estimation of parameters of stock condition (Maunder et al., 2014; Sampson, 2014).

4.4. Conclusions and outlook

A general pattern of declining stocks has not only been presented by reports made by Colombia's fisheries authority in the past years, but it is the perception shared by different stakeholders in the country. Extensive interviews carried out with fishers, community leaders in coastal areas and fisheries experts from government and academic institutions between 2008 and 2009, revealed a general perception of declining fishing resources and of increasing fishing effort in the country's small-scale fisheries (Saavedra-Diaz, 2012).

Despite the limitations to adequately estimate the indicators of stock condition based on length-frequency data only, most of our results are consistent with those obtained by previous assessments in Colombia and neighboring countries for the three selected target species but, moreover, highlight the value that length-based methods still have to assess stocks in data-poor contexts around the world where modern population dynamics models or integrated approaches cannot be applied due to lack of available data (Cope and Punt, 2009).

Our study shows how differences in sampling schemes to collect LFCD in a fairly delimited coastal zone can impose different degrees of uncertainty to data analyses and, more importantly, may even lead to different management conclusions. A higher uncertainty was observed in estimates derived from NGO data, which we attribute to a very specific and selective type of fisheries in the northern sub-region of the Pacific coast. While GOV data included a wider range of fleets and gears (e.g. gillnets of different mesh sizes and other type of nets) that are used along the entire coast, NGO data was dominated by catches made with hooks in more offshore pelagic or deeper environments, given by the particular environmental characteristics of that sub-region. In this sense, our estimated VBGP, as well as their related E and F values, could be biased due to using non-corrected LFCD that did not account for the intrinsic selectivity of the fisheries (Taylor and Martell, 2005). In order to improve future stock assessments and reduce the underlying uncertainty of LFCD in multi-gear and small-scale fisheries settings, a stratified random sampling should be designed (Sparre and Venema, 1998) based on a previous assessment of current fishing effort and how it changes spatially, temporally and among the different gears (McCluskey and Lewison, 2008). In the case of Colombia, this change would not necessarily mean additional costs compared to the current sampling scheme but a redistribution of the current sampling effort.

On the other hand, taking into account the large influence that the selectivity of the fisheries has on the estimation of VBGP, natural mortality and fisheries mortality parameters, specific research on estimates of gear selectivity is highly recommended to be able to correct LFCD before conducting analyses. Selectivity studies should differentiate not only among main type of gears but also among the more common mesh sizes and hook sizes that are used by fishers (Millar and Fryer, 1999; Millar and Holst, 1997). A complementary analysis of the influence of different fishing grounds on the size distribution of the catch will also help to refine stock assessments and their interpretation (Punt et al., 2014; Sampson, 2014).

Despite the appealing nature of LBI to managers and other stakeholders, due to their ease of calculation and interpretation (i.e.: no need for complex calculations of growth and mortality parameters), caution must be taken to draw conclusions out of "snap-shot" data or when estimated values of *Lm* do not correspond to the stock assessed. Even though the *Pobj* indicator and the decision-tree used here (Cope and Punt, 2009) infers and takes into account a selectivity pattern of the fisheries, improvements in the sampling scheme as those described above and acquiring longer time series data will increase reliability of results of stock assessments based on LBI. Participative monitoring in rural coastal

Table 7
Estimated values for length at maturity (*Lm*) and proportion of mature fish in the catch (*Pmat*) for *S. sierra, L. guttatus* and *B. clarkae.* Case where studies were performed separately for female (F) or male (M) fishes are indicated. All values refer to total length, except where indicated ([a]).

Species	*Lm* (cm)	*Pmat*	Source	Country
S. sierra	58.80	0.15	Present study (GOV)	Colombia
	58.80	0.38	Present study (NGO)	Colombia
	58.50	0.05	AUNAP and UNIMAGDALENA (2013b)	Colombia
	57.50		Barreto and Borda (2008)	Colombia
	58.9 (F)		Ordoñez et al. (2017)	Colombia
	44.3[a] (F)	0.3	Aguirre-Villaseñor et al. (2006)	Mexico
	56.4 (M)	0.26–0.29	Lucano-Ramírez et al. (2011)	Mexico
	59.3 (F)	0.26–0.29	Lucano-Ramírez et al. (2011)	
L. guttatus	35.30	0.47	Present study (GOV)	Colombia
	35.30	0.56	Present study (NGO)	Colombia
	41.3	0.17	AUNAP and UNIMAGDALENA (2013b)	Colombia
	30.6	0.39	Sarabia-Méndez et al. (2010)	Mexico
	33		Rojas (1997)	Costa Rica
B. clarkae	75.40	0.48	Present study (GOV)	Colombia
	75.40	0.51	Present study (NGO)	Colombia
	73	0.49	AUNAP and UNIMAGDALENA (2013b)	Colombia
	71		Muñoz (1999)	Colombia
	62.3		Acevedo et al. (2007)[b]	Colombia
	71.9	0.13	Herrera et al. (2016)	Costa Rica

[a]Furcal length used, instead of total length.
[b]Based on data collected from 1994 to 1996.

areas could greatly improve landing sites coverage and help the implementation of the new sampling design (Díaz-Merlano et al., 2016; Ramírez et al., 2017), but overall control and supervision from the fisheries authority is recommended so that there is a single handler and curator of the database. Finally, tropical developing countries, like Colombia, will also greatly benefit from investing in acquiring fisheries-independent data to increase the knowledge about the status of the populations and their life cycles. This could be a joint effort with neighboring countries that share the use and management responsibilities of the fished stocks.

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Conflict of interest

None.

Appendix A. Supplementary data

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